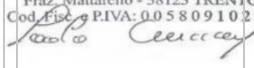


UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIS 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

IDENTIFICATION TABLE

Client	PBDiM - Energopol
Project	Udrożnienie Łódzkiego Węzła Kolejowego (TEN-T), Etap II, Odcinek Łódź Fabryczna - Łódź Kaliska/Łódź Żabieniec – Projekt POIIS 5.1-15
Task	TBM 13.08 m - CONSTRUCTION FOLLOW-UP
Type of document	Daily Report 12/09/2024
Date	13/09/2024
Document ID	LWK-000-T-KO-SWS-210483_00.docx
Language	English
Number of pages	15

AUTHOR		REV
Opracował	Miriam Polizzi	00
Projektant	Paolo Cucino - Inżynierska mostowa i konstrukcyjno-budowlana, upr. nr ŁOD/TR/0247/19	SWS Engineering S.p.A. Via della Stazione, 27 38123 Trento Cod. Fisc. 00580910222 P.IVA: 00580910222 

INDEX

1.	INTRODUCTION	3
2.	FACE PRESSURES AND EXCAVATION VOLUMES	4
3.	THRUST AND CUTTERHEAD TORQUE	6
4.	GROUT INJECTION	7
5.	GUIDANCE	8
6.	MONITORING	9
7.	COMPENSATION GROUTING ACTIVITIES	14
8.	SOIL CONDITIONING	15

1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

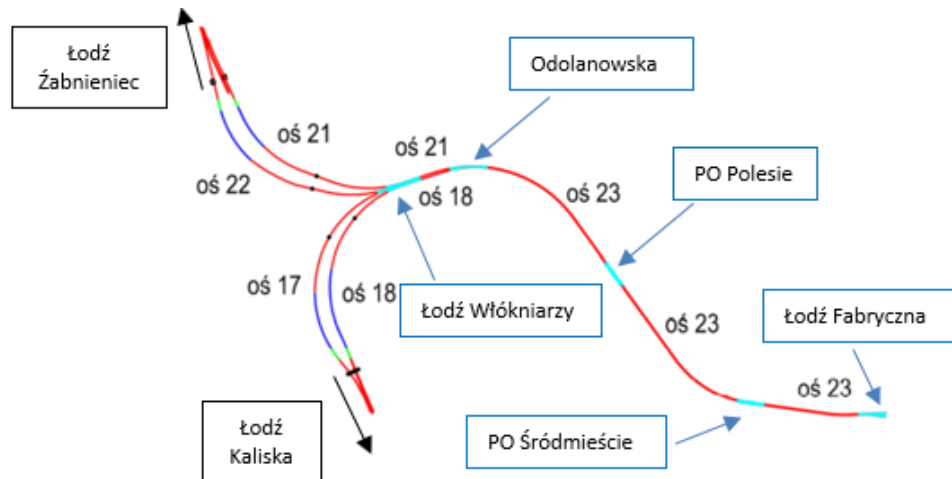


Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 12/09/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+628.92 after the erection of ring n. 269; which corresponds to a total excavation length of 337.55 m from the starting chainage.

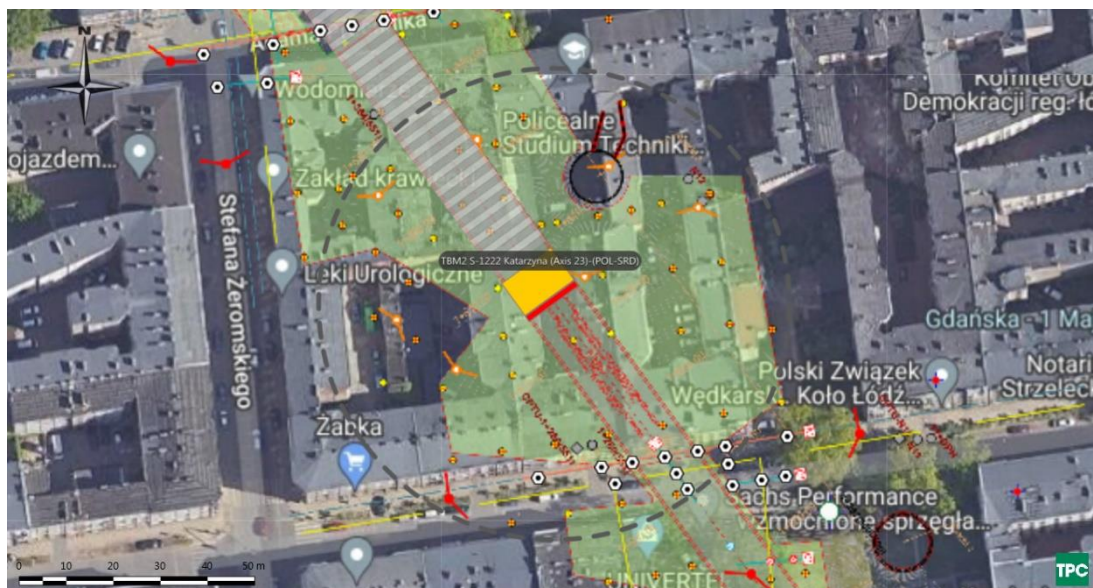


Figure 1-2 Plan view of TBM Position on 13/09/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.8 bar. The value is within the attention values.

Figure 2-1 shows EPB sensors.

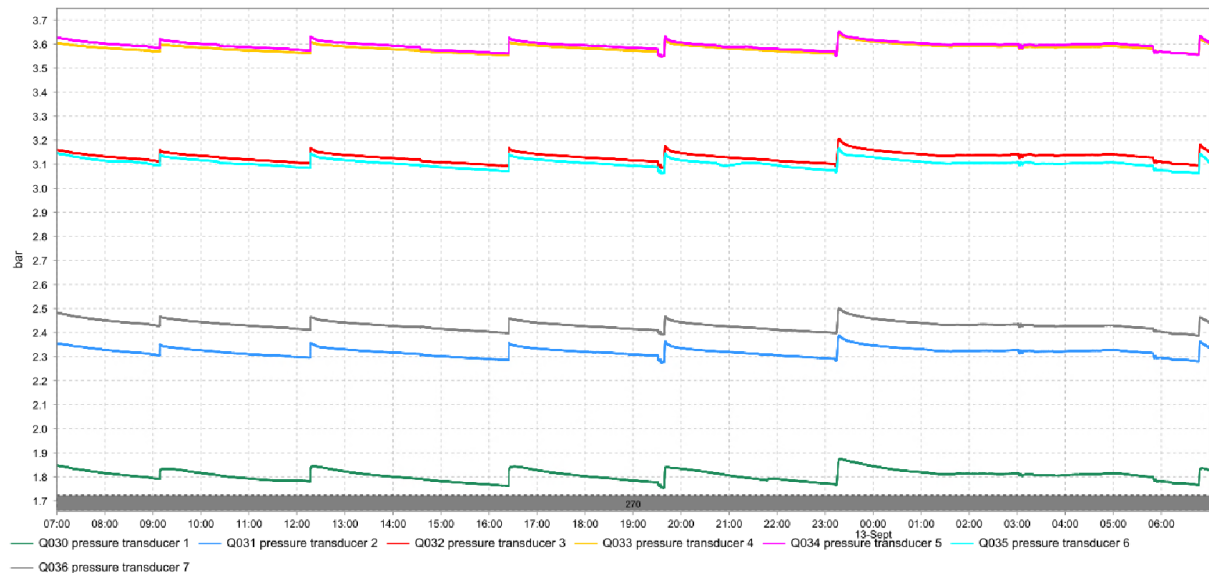


Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.5 bar indicating that the bulk chamber is filled. The observed value in delta between pressure transducer 1 and 2 is most likely related to the injection of bentonite into the excavation chamber.

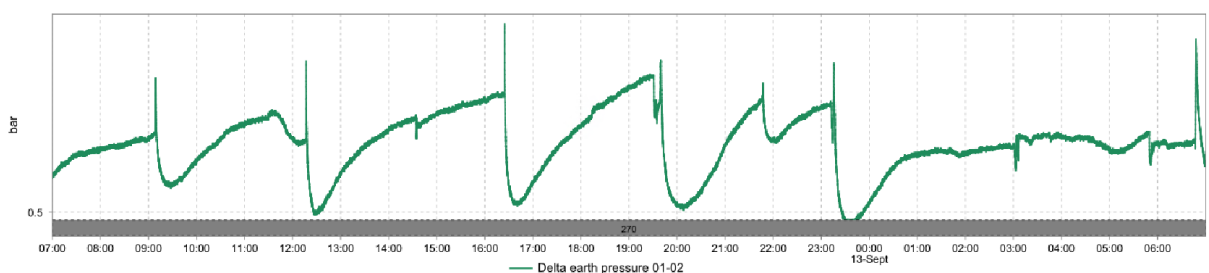


Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 shows values lower than n. 1. It could be necessary the recalibration of belts before the restart of TBM excavation. The results of Belt scales have to be compared at the end of excavation of the ring. The belt scale value is a raw indication of the excavated material quantity.

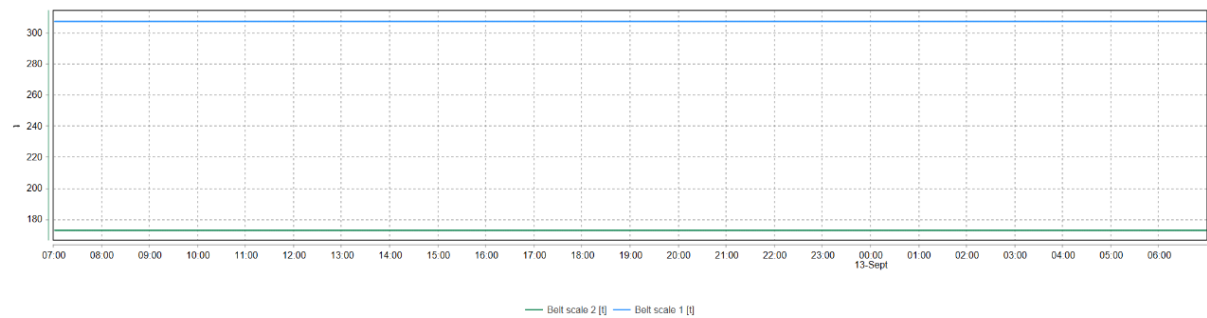


Figure 2-3 Belt scales

3. THRUST AND CUTTERHEAD TORQUE

Graph for torque, advance rate and thrust force are below shown. The torque graph shows non-zero values due to the rotation of the cutterhead.

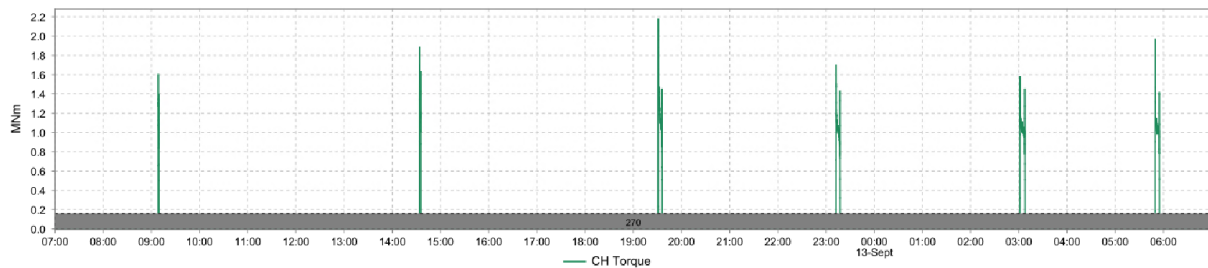


Figure 3-1 CH Torque

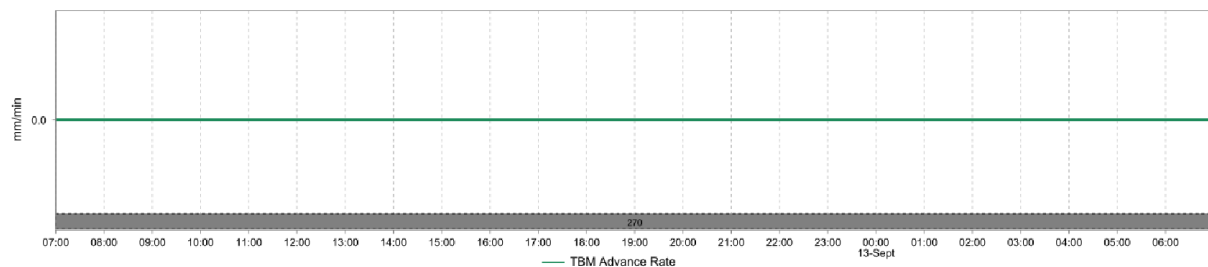


Figure 3-2 Advance rate

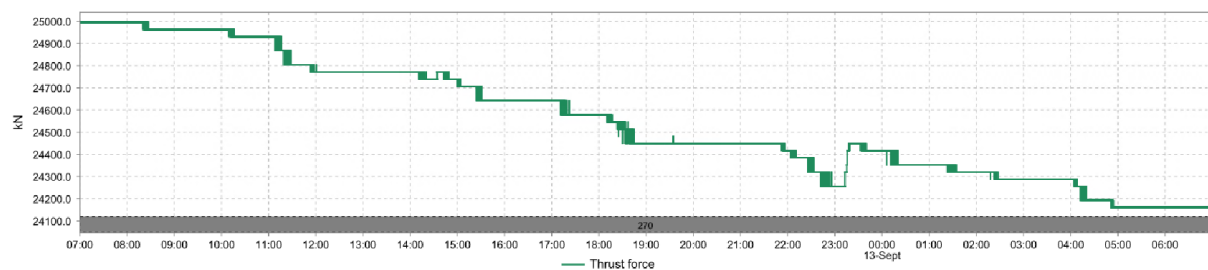


Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection volume has to be compared to minimum theoretical annular volume $\pm 10\%$ of 13.54 m^3 at the end of ring excavation. Ratio B/A is at 5%. It was noted that peaks observed for average injection pressure lines are related to cleaning processes.

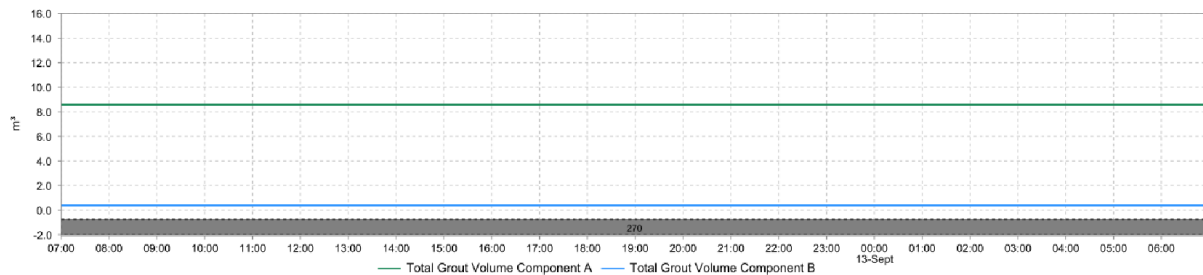


Figure 4-1 Grout volumes

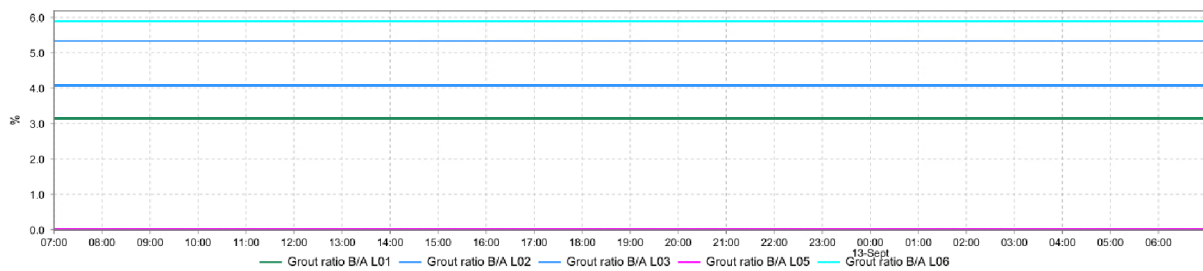


Figure 4-2 Volume ratio Components B/A

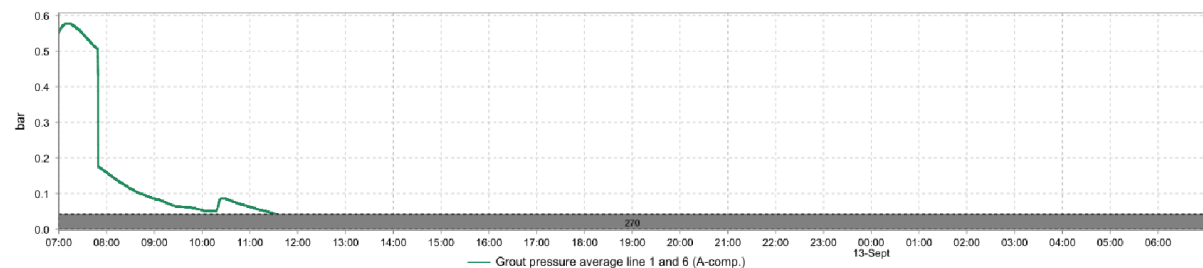


Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about -15.85 mm on the front edge of the shield and the horizontal deviation of about -17.86 mm (Figure 5-1). Maximum radial deviation is of 23.88 mm.

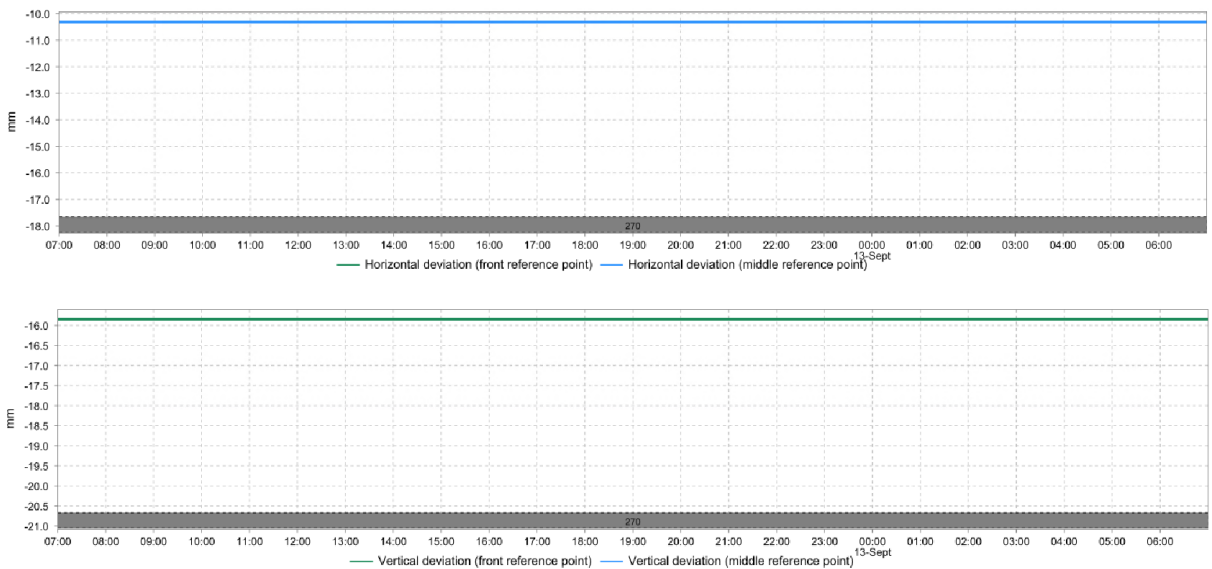


Figure 5-1 Machine deviation from the alignment

6. MONITORING

Based on data from Geoscope (manual geodetic measurements) and Beyond (hydromonitoring measurements), building subsidence has been observed but is showing signs of uplift.



Figure 6-1 Monitoring points – Geoscope

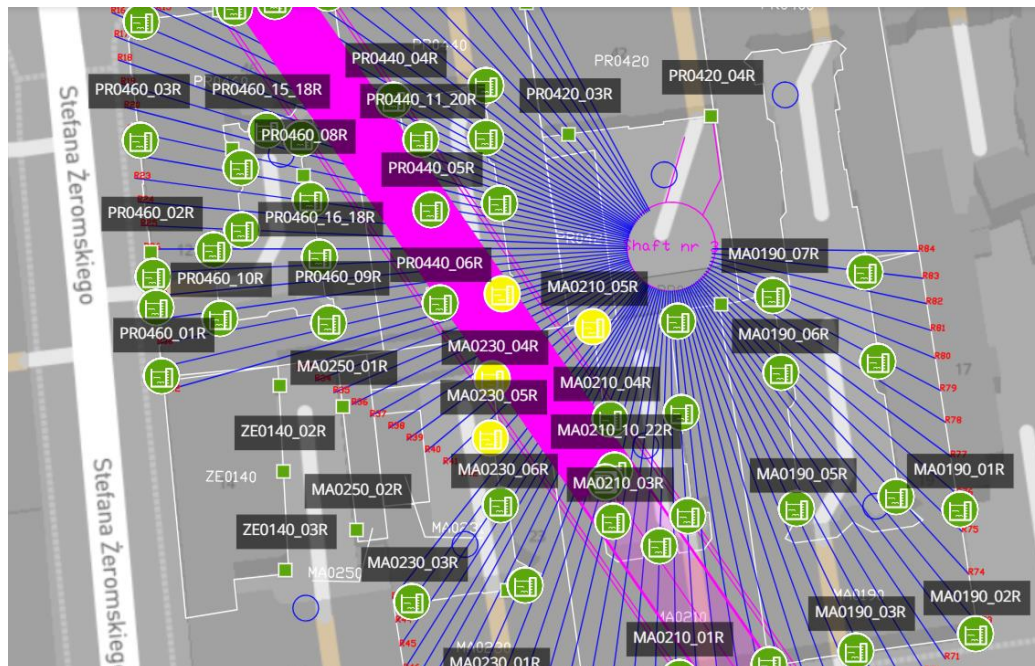


Figure 6-2 Monitoring points – Beyond

On PR0440, Geoscope monitoring points show uplift trends, especially for PR0440_06R and PR0440_07R. In Beyond, monitoring points have shown stabilizing trends since 9:00 pm on 12/09/2024, following the prior uplift.

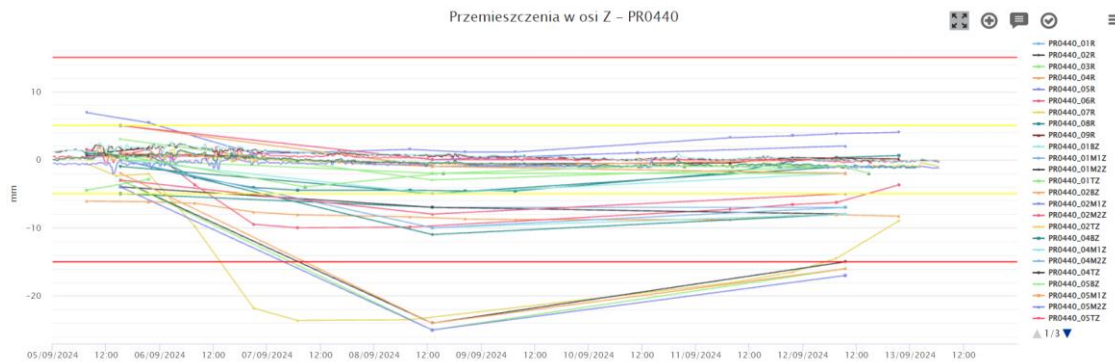


Figure 6-3 Settlement of building PR0440 - Geoscope



Figure 6-4 Settlement of building PR0440 - Beyond

On PR0460, from Geoscope most of the monitoring points are stable, except for PR0460_08R and PR0460_09R which show a small uplift (for PR0460_09R exceeding attention limit). However, not all trends can be fully assessed, as some points have not recorded new measurements since the morning of 10/09/2024. In Beyond, monitoring points are stable.

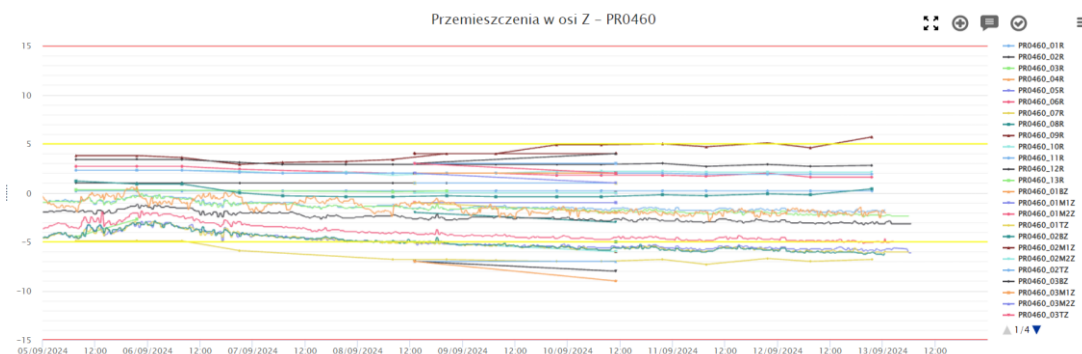


Figure 6-5 Settlement of building PR0460 - Geoscope

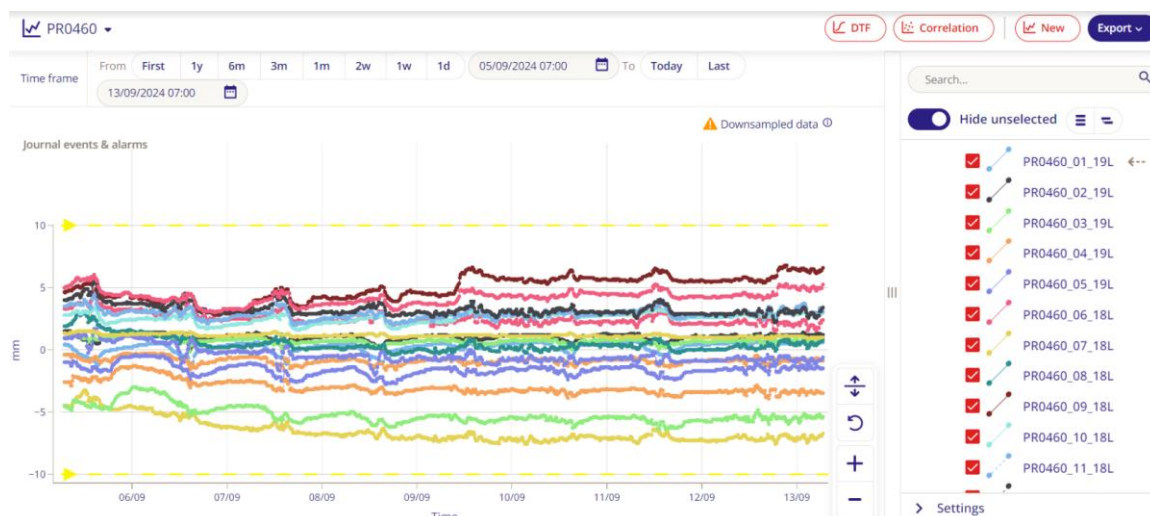


Figure 6-6 Settlement of building PR0460 - Beyond

On MA0190, monitoring points are stable.

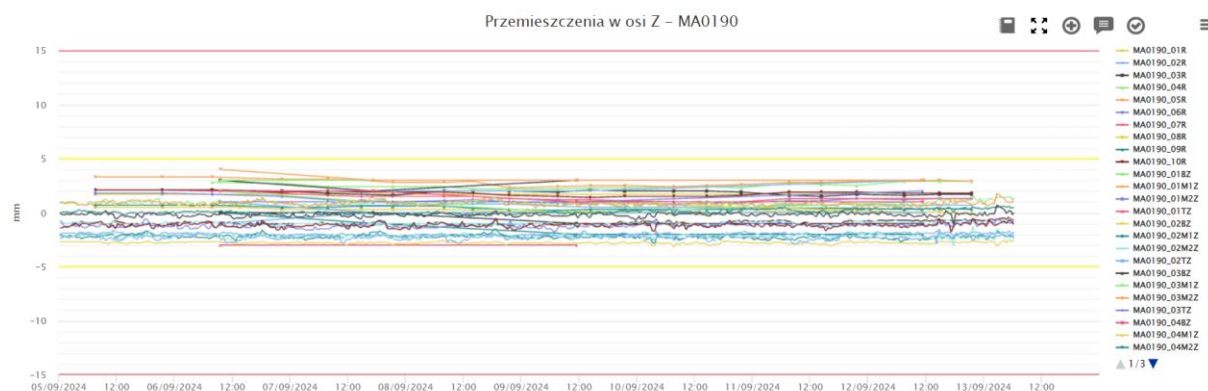


Figure 6-7 Settlement of building MA0190 - Geoscope



Figure 6-8 Settlement of building MA0190 - Beyond

On MA0210, Geoscope monitoring points MA0210_04R, MA0210_05R and MA0210_06R show uplift trends. In Beyond, monitoring points MA0210_03_22L, MA0210_04_22L, MA0210_05_22L show uplift trends.



Figure 6-9 Settlement of building MA0210 - Geoscope

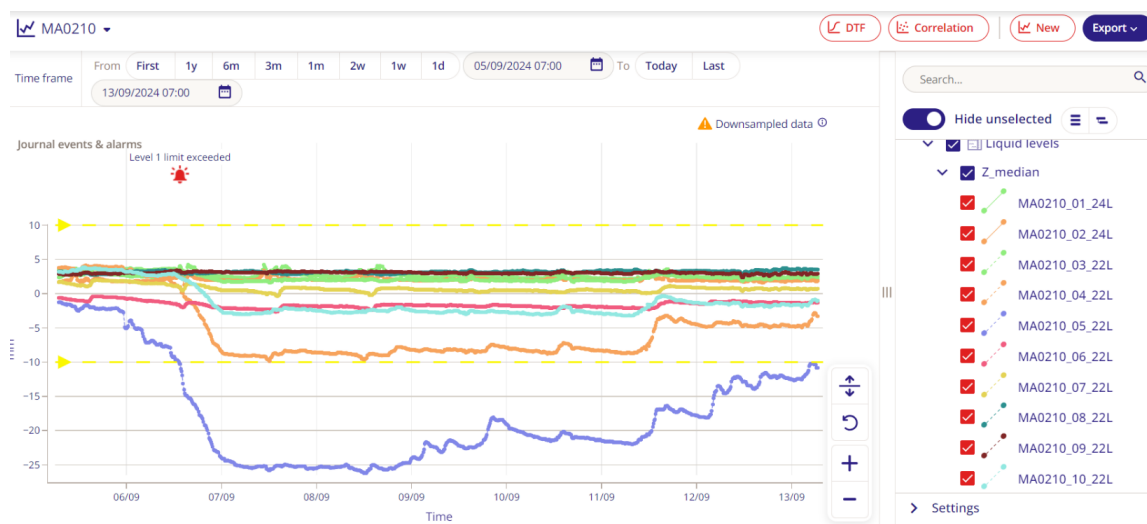


Figure 6-10 Settlement of building MA0210 - Beyond

On MA0230, Geoscope monitoring points MA0230_05R, MA0230_06R and MA0230_07R show small uplift trends. In Beyond, monitoring points MA0230_05_21L, MA0230_06_21L show small uplift trends.

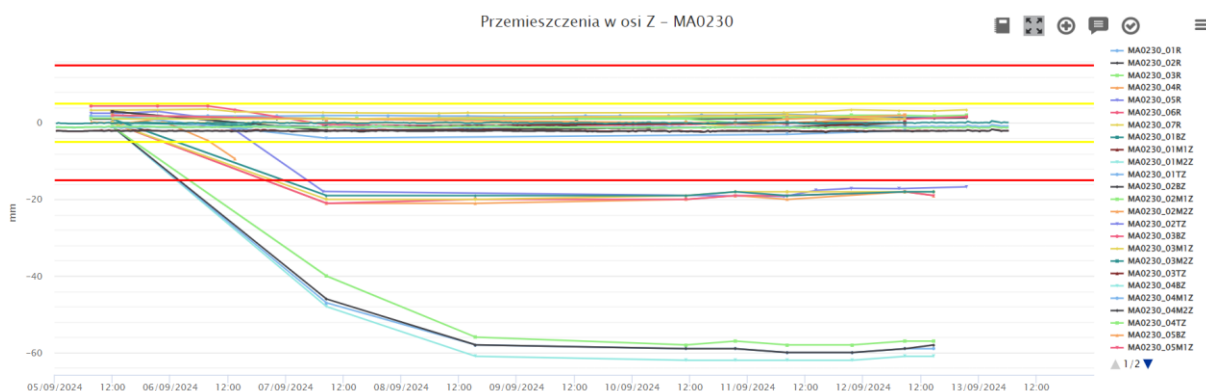


Figure 6-11 Settlement of building MA0230 - Geoscope

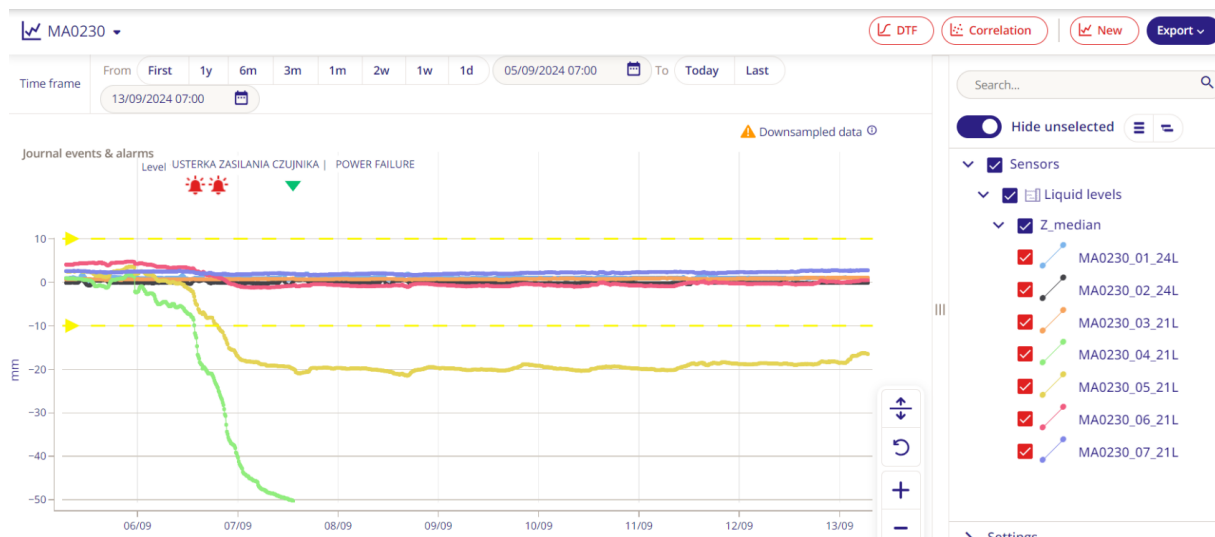


Figure 6-12 Settlement of building MA0230 - Beyond

On MA0250, monitoring points are stable, except for MA0250_02R which shows a small uplift trend.

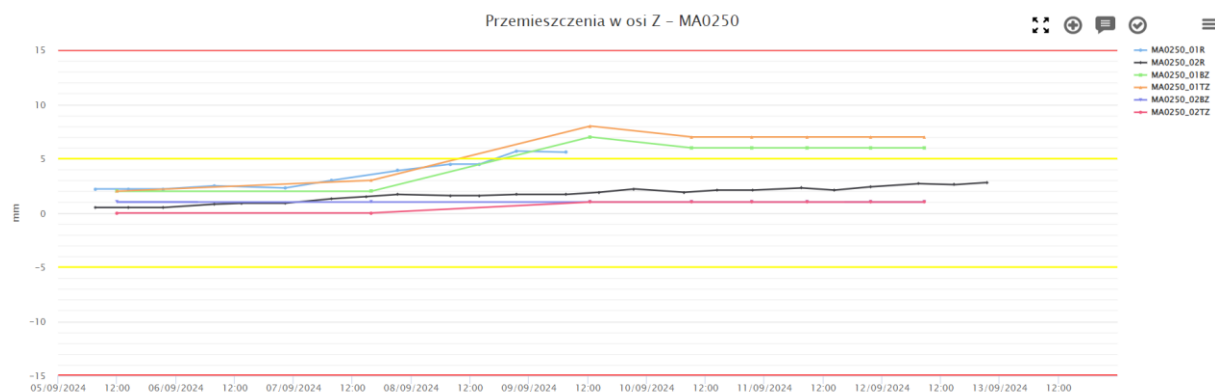


Figure 6-13 Settlement of building MA0250 – Geoscope

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 12/09/2024 of Soletanche, compensation grouting has been performed on the 12/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 12349.52 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure.

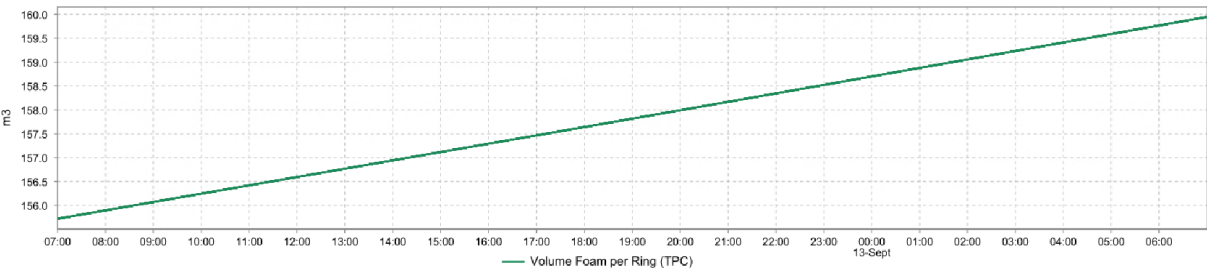


Figure 8-1 Injected foam volume

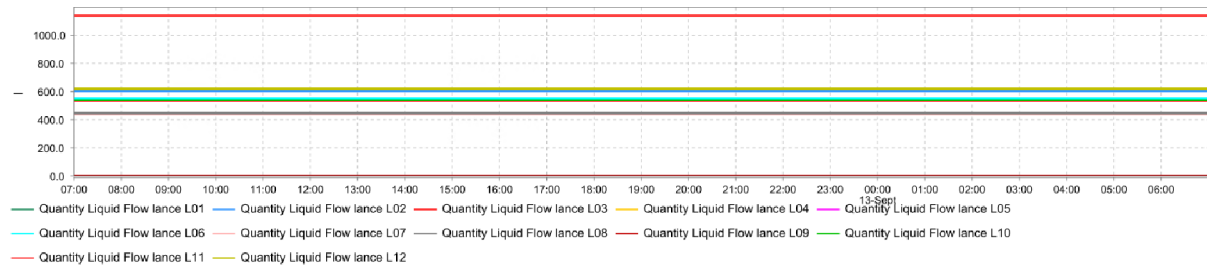


Figure 8-2 Conditioning liquid volume